Md Mridul Hossain

GenAI Developer | AI Engineer | Software Engineer

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Summary

GenAI developer and software engineer with 2+ years of professional experience, building an autonomous **multi-agent LLM platform** at Deets Software Ltd, with an M.Eng. in Software Engineering from Concordia University. Hands-on with **LangGraph** orchestration, **RAG** on PostgreSQL + **pgvector**, prompt engineering across **Claude** and **GPT-4o**, and Python backends with **FastAPI**, Celery, and Redis, backed by a full-stack and enterprise-software foundation.

Skills

Generative AI: LLM application development, multi-agent systems, LangGraph, RAG, vector embeddings, semantic search, prompt engineering, Anthropic Claude, OpenAI GPT-4o, human-in-the-loop (HITL) guardrails
Backend & Data: Python, FastAPI, SQLAlchemy, Celery, Redis, PostgreSQL, pgvector, MySQL, Oracle DB, MS SQL Server, REST & SOAP APIs, JSON, XML
Frontend & Languages: Next.js, React, TailwindCSS, JavaScript, PHP (Laravel), Java, SQL, HTML5, CSS3, C++
Tools & Practices: Git, Linux, Bash, asynchronous task queues, system design, requirement gathering

Professional Experience

GenAI Developer **June 2025 – Present**
Deets Software Ltd

- Building **AutoMarket AI**, an Agent-as-a-Service platform that acts as an autonomous "digital CMO" for SMBs and marketing agencies, automating cross-channel campaign planning, content creation, and performance optimization.
- Architecting a **5-agent LangGraph system** (Orchestrator/CMO, Content & SEO, Paid Media, Retention, Analytics & CRO) with typed shared state, parallel execution for ad and email workflows, and a conditional feedback loop that sends underperforming KPIs back to the orchestrator.
- Developing a **RAG** brand-memory layer that chunks and embeds brand guidelines, campaign history, and product catalogs into **PostgreSQL + pgvector** for cosine-similarity retrieval, keeping every agent output on-brand.
- Engineering the **FastAPI** and SQLAlchemy backend, with **Celery + Redis** running long-lived agent workflows asynchronously, and routing prompts across **Claude** (strategy) and **GPT-4o** (structured output).
- Implementing **human-in-the-loop** guardrails: any agent action that spends budget or publishes content is intercepted into an approval queue and reviewed through Approve / Reject / Revise endpoints and a Next.js dashboard.

Software Engineer **February 2023 – October 2023**
Uttara University

- Re-engineered the university's web-based ERP platform with **PHP (Laravel)**, HTML, CSS, and Bootstrap, modernizing day-to-day institutional operations.
- Migrated roughly **370,000 records** from MS SQL to MySQL with zero data loss, validating integrity through direct database queries.
- Built bulk ID-card processing and integrated secure payment gateways with banking partners.

Junior Backend Developer **January 2023**
Techvillage

- Contributed backend features and code reviews to Paymoney, a fintech product, in a shared **Git** codebase.
- Integrated the **Flutterwave** payment gateway over **REST APIs**, parsing and validating JSON responses and testing currency-exchange endpoints.

Projects

Context-Aware Travel Companion *Python, KivyMD, OWL, SQLite | Concordia University – Fall 2025*

- Built a **Python (KivyMD)** app that recommends places from live weather, time of day, and user preferences by reasoning over an **OWL ontology** (Owlready2), using the **Google Places** and **OpenWeather** APIs, Google OAuth, and SQLite.

Education

Concordia University — *Master of Engineering, Software Engineering* *January 2024 – August 2026*
American International University-Bangladesh — *B.Sc. in Software Engineering* *September 2018 – March 2022*